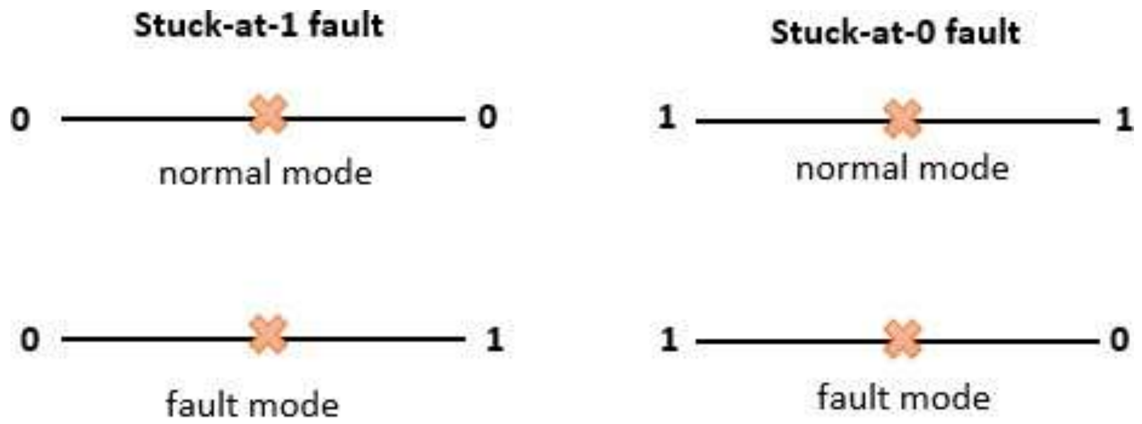
Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Digital VLSI Testing (01CT0629)	**Aim: To model and detect stuck-at-0 and stuck-at-1 faults in a combinational circuit using ATPG (Automatic Test Pattern Generation).**	
Experiment No: 5	**Date:04-03-2026**	**Enrolment No:92301733056**

AIM: To model and detect stuck-at-0 and stuck-at-1 faults in a combinational circuit using ATPG (Automatic Test Pattern Generation).

IDE: EDA Playground

THEORY:

A Stuck-At Fault (SAF) is one of the most common fault models. A signal is considered stuck-at-0 (sa-0) or stuck-at-1 (sa-1) when it is permanently tied to logic 0 or 1, regardless of the actual circuit function.



i.e.: If a wire or net is stuck-at-0, it always reads as 0—even when the logic should make it 1

Stuck-at faults occur due to:

- Manufacturing defects (broken interconnects, bridging, open connections)
- Oxide defects in gates
- Short circuits in layout

The Automatic Test Pattern Generation (ATPG) technique is used to generate inputs that can detect such faults by

- Sensitizing the fault site
- Propagating the fault to an observable output
- Justifying internal signals

In this experiment, we simulate a stuck-at fault at input B[1] of the 4-bit ripple carry adder. Instead of connecting it to B[1], we tie it permanently to '0' to represent a stuck-at-0 fault.

SYSTEM ARCHITECTURE:

The system consists of

- Full Adder Module: Basic building block that performs 1-bit binary addition.
- Ripple Carry Adder: Four full adders connected in sequence to perform 4-bit addition
- Fault Injection: Input B[1] is forced to logic 0, simulating a stuck-at-0 fault.
- Testbench: Applies ATPG vectors and logs Sum, Carry, and Cout outputs.

DESIGN AND IMPLEMENTATION:

A. Full Adder Logic

Implements sum = $A \oplus B \oplus Cin$ and carry = AB + BCin + ACin.

B. Ripple Carry Adder

Connects 4 full adders such that:

- FA0: A[0], B[0], Cin
- FA1: A[1], 1'b0 (B[1] stuck at 0), Carry0
- FA2: A[2], B[2], Carry1
- FA3: A[3], B[3], Carry2 → Cout

VERILOG CODE:

A. Full adder Module

```
module full_adder( input logic
          a, b, cin, output logic
          sum, cout
); assign sum = a ^ b ^ cin;
          assign cout = (a & b) | (b & cin) | (a & cin);
endmodule
```

B. Ripple Carry Adder Module module

```
ripple_carry_adder_4bit(
input logic [3:0] A, B, input
logic        Cin, output logic
[3:0] Sum, output logic
Cout
);
          logic [2:0] carry;
```

```
        full_adder FA0 (A[0], B[0], Cin,      Sum[0],        carry[0]);
        //FA1 with B[1] stuck-at-0 fault injection:
        full_adder FA1 (A[1], 1'b0, carry[0], Sum[1], carry[1]); // B[1] sa-0
        // To inject stuck-at-1 instead, replace 1'b0 with 1'b1 here. full_adder
        FA2 (A[2], B[2], carry[1], Sum[2], carry[2]); full_adder FA3 (A[3],
        B[3], carry[2], Sum[3], Cout);
endmodule
```

C. Testbench `timescale

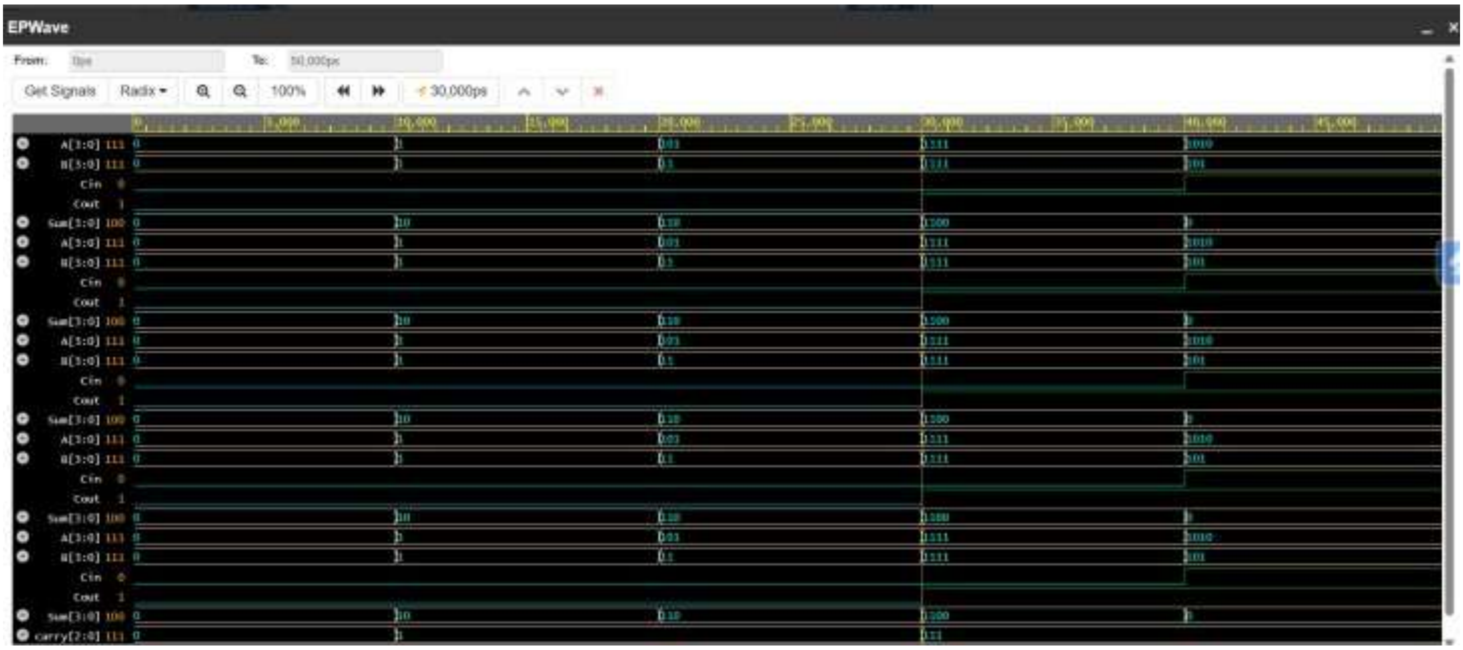
```
1ns/1ps module tb; logic
[3:0] A, B;
        logic      Cin;
        logic [3:0] Sum; logic
            Cout;

        // Instantiate Device Under Test
        ripple_carry_adder_4bit uut (A, B, Cin, Sum, Cout);

        // Generate VCD waveform initial
        begin
                $dumpfile("dump.vcd");
                $dumpvars(0, tb); end
initial begin
                // ATPG-style vectors
                A = 4'b0000; B = 4'b0000; Cin = 0; #10;
                A = 4'b0001; B = 4'b0001; Cin = 0; #10;
                A = 4'b0101; B = 4'b0011; Cin = 0; #10;
                A = 4'b1111; B = 4'b1111; Cin = 0; #10;
                A = 4'b1010; B = 4'b0101; Cin = 1; #10;
                $finish; end
endmodule
```

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Digital VLSI Testing (01CT0629)	**Aim: To test parity generator and checker for bit error detection.**	
Experiment No: 05	**Date:04/03/2026**	**Enrolment No: 92301733056**

SIMULATION RESULTS:



CONCLUSION:

In this experiment, a 4-bit Ripple Carry Adder was designed using full adder modules, and stuck-at-0 and stuck-at1 faults were injected at a selected input line. ATPG-style test vectors were applied through simulation to observe the behavior of the circuit. The faulty outputs were compared with the expected normal outputs to detect the presence of faults. The results showed that specific input combinations are required to activate and propagate faults to the output.

Thus, the experiment demonstrates the importance of fault modeling and Automatic Test Pattern Generation (ATPG) in ensuring reliability and correctness of combinational VLSI circuits.